

Ehsan Moradi

Curriculum Vitae

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Education

- Sept 2023 – **M.Sc. in Computer Engineering**, GPA: 3.77/4, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
- Present
- **M.Sc. Thesis:** "A Proactive Container-based Auto-scaling Approach for a Hierarchical Orchestration Framework for Edge Computing," under supervision of Dr. S.A. Javadi
- Sept 2017 – **B.Sc. in Computer Science**, GPA: 3.07/4, IK International University, Qazvin, Iran
- Sept 2022
- **B.Sc. Project:** "Full-stack web-application and mobile app of an online and interactive quiz game (called *QuestionParty*)," under supervision of Dr. Ali Khaleghi

Professional Experience

- Feb 2022 – **Software Engineering Intern**, DiSL (Data in Science Laboratory), Qazvin, Iran
- Mar 2023
- Served as the core developer for "QuestionParty," an interactive, cross-platform application.
 - Architected and developed scalable backend RESTful APIs and real-time communication features using Python, Flask (SocketIO, Login), and MongoDB (PyMongo).
 - Built responsive user interfaces for both mobile and web platforms using Flutter and Flutter Web.
 - Managed server deployment and containerization utilizing Docker, Nginx, uWSGI, and Linux environments to ensure high availability and optimal performance.

Publications

- Moradi, E., & Javadi, S. A., "Vicinity-Aware Proactive Auto-Scaling for Containerized Edge Clusters," *Manuscript in preparation*. Code repository: GitHub.

Honors & Awards

- Ranked 2nd among all M.Sc. students in the Department of Computer Engineering (Class of 2023).
- Ranked among the top 1% of participants in Iran's university entrance exam for M.Sc..
- Ranked among the top 2% of participants in Iran's university entrance exam for B.Sc..

Research and Engineering Projects

- **Distributed Systems (M.Sc.)**
Implemented distributed synchronization and analyzed consistency models and Paxos-based transaction protocols.
- **Applied Machine Learning (M.Sc.)**
Reproduced a 2024 Computer Vision research paper using a hybrid Deep Learning model (VGG19 + SVM).
- **Structure Assembling Robot (Cross-Disciplinary Prototype)**
Engineered microcontroller embedded logic and a native Android app to wirelessly control a mechanical assembly robot via Bluetooth, contributing to the prototype's US Patent Application (US20240084610A1).
- **Systems Analysis and Design**
Conducted comprehensive process modeling and system analysis for the "South Tehran Health Center" organization.

Teaching Experiences and Activities

- Spring 2026 **Teaching Assistant**, Two Courses: Applied Machine Learning and Distributed Systems, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
- Fall 2025 **Teaching Assistant**, Logic Circuit Design, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
- Spring 2025 **Seminar Presenter: Recurrent Neural Networks & LSTMs**, Applied Machine Learning, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

Related Courses

Distributed Systems (4/4) • Cloud Computing (4/4) • Applied Machine Learning (4/4) • Dependable Systems Design (4/4) • Systems Analysis and Design (4/4)

Skills

- **Distributed & Cloud:** Kubernetes, Docker, Hadoop (HDFS/YARN), MapReduce, Firebase (BaaS), Microservices.
- **AI & Machine Learning:** Python, Deep Learning (CNN, RNN, LSTM), Transfer Learning, NLP, Bayesian Networks.
- **Development & Databases:** Flutter, Flask, MongoDB, Firestore, RESTful APIs, Git.
- **Networking:** Network+, CCNA, MCSA (Certified).

Languages

- **English:** Professional (Certified)
TOEFL iBT: Overall 6.0/6 (CEFR: C2) — Comparable to 118/120
Sections: Reading 5.5, Listening 6.0, Speaking 5.5, Writing 6.0
- **Persian:** Native